Course Name: Systems & Signals Processing	**Date:** May 19, 2008
Course No: 402-322	**ExamTime:** 50 min
Instructor: Eng. Juma Abu-Ghalune	**No. of pages:** 1

[redacted]

Answer only 4 problems from the 5 problems below.

1. Find the energy spectral density of the signal $x(t) = 8sinc(10t)$ then find the total energy of this signal.

2. Consider the signal $x(t) = 2 + 6.\cos^2(150\pi t) + \sin(100\pi t)\cos(200\pi t)$. Find:
   a. The fundamental period.
   b. The Fourier series coefficients.
   c. The average power.

3. Consider a fixed linear system with amplitude response $3.\Pi[w/600\pi] + 3.\Pi[w/400\pi]$ and phase response $-0.002\,w$ for $|w| \leq 250\pi$, 0.5π for $w < -250\pi$, and -0.5π for $w > 250\pi$. Plot the amplitude and the phase of the transfer function of the system then find the output and determine the type of distortion for the following inputs:
   a. $x_1(t) = 2.\cos(225\pi t) + 4.\cos(275\pi t)$.
   b. $x_2(t) = 4.\cos(150\pi t) + 2.\cos(225\pi t)$.

4. Consider a periodic signal $x(t)$ is defined in the fundamental period $-2 < t < 2$ by $x(t) = 2t$, for $|t| < 1$, and $x(t) = 0$, for $1 < |t| < 2$. Find:
   a. The Complex Fourier series of $x(t)$.
   b. X_n, a_n, and b_n for $n < 5$.
   c. The average power contained in the first four harmonics.

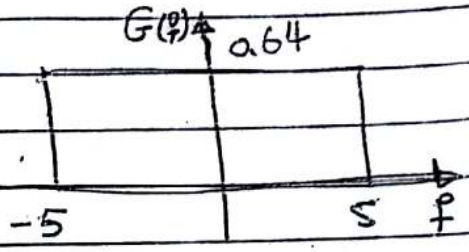
5. Using the integral definition form find the :
   a. Fourier transform of $x(t) = 2.\exp(0.2t).\delta(t - 0.5)$.
   b. Inverse Fourier transform of $Y(f) = 4.\exp(0.5f).u(2 - f)$.

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Q.1 $x(t) = 8\,\text{sinc}(10t) = 0.8\,(10\,\text{sinc}(10t))$

$X(f) = 0.8\,\Pi\left(\frac{f}{10}\right)$.

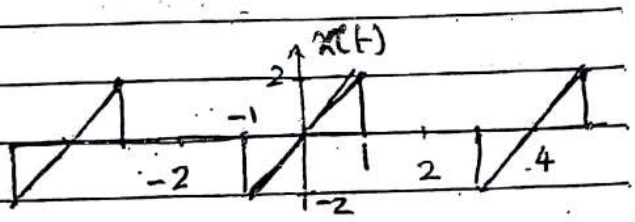
$G(f) = |X(f)|^2 = 0.64\,\Pi\left(\frac{f}{10}\right)$



$E = 2\int_0^{\infty} G(f)\,df = 2(0.64)\int_0^{5} df$

$= 6.4$ J.

Q.4 $x(t) = \begin{cases} 2t & -1 < t < 1 \\ 0 & 1 < |t| < 2 \end{cases}$



a $X_n = \frac{1}{4}\int_{-1}^{1} 2t\, e^{-j2\pi \frac{n}{2} t}\,dt = \frac{2}{2}\int_0^1 t\, e^{-j2\pi f_0 t n}\,dt$

$= \frac{t}{-j2\pi n f_0} e^{-j2\pi n f_0 t}\Big]_0^1 - \frac{1}{(j2\pi n f_0)^2} e^{-j2\pi n f_0 t}\Big]_0^1$

$= j\frac{2}{n\pi}\left[e^{-j2\pi n f_0} - \right] + \frac{4}{n^2\pi^2}\left[e^{-j2\pi n f_0} - 1\right]$

$= -\frac{4}{n^2\pi^2} + \frac{4}{n^2\pi^2} e^{-j\frac{n\pi}{4}} + j\frac{2}{n\pi} e^{-j\frac{n\pi}{4}}$

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Q.5 a. $x(t) = 2e^{0.2t}\,\delta(t-0.5)$.

$$X(f) = 2\int_{-\infty}^{\infty} e^{(0.2+j2\pi f)t}\,\delta(t-0.5)\,dt$$

$$= 2e^{0.5(0.2-j2\pi f)} = 2e^{(0.1+j\pi f)}$$

b. $Y(f) = 4e^{0.5f}\,u(2-f)$

$$y(t) = 4\int_{-\infty}^{2} e^{(0.5+j2\pi t)f}\,df = \frac{4}{[0.5+j2\pi t]}\,e^{2(0.5+j2\pi t)}$$

$$= \frac{4}{[0.5+j2\pi t]}\cdot e^{(1+j4\pi t)}$$

Q.2 $x(t) = 2 + 6\cos^2(150\pi t) + \sin(100\pi t)\cos(200\pi t)$

$= 5 - 0.5\sin(100\pi t) + 3\cos(300\pi t) + 0.5\sin(300\pi t)$.

a. $T_0 = \frac{2\pi}{100\pi} = 0.02\,s$

b. $X_0 = 5$; $X_1 = +j\frac{1}{4}$, $X_3 = 1.5 - j0.25$

c. $P_{avg} = 25 + 2\left[(0.25)^2 + (1.5)^2 + (0.25)^2\right]$

$= 29.75\,W$.

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Al Isra Private University
Faculty of Engineering
Electrical Engineering Department

Student Name:_____
Student Number:_____

Second Semester: 2005/2006
Second Exam

Course Name:	Systems & Signals Processing	Date:	May 7, 2006
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $x(t) = 5.exp(-(t+0.1)^2)$.
   b. $y(t) = 4.sin(50\pi t).cos(500\pi t)$.
   c. $z(t) = 4/(1 - j\pi t)^2$.

2. Using Table 4-1 and Table 4-2, find the Fourier inverse transform of:
   a. $X(f) = 4exp(-4f)u(f)$.
   b. $Y(f) = (-j/\pi f)exp(j0.2\pi f)$.

3. The complex exponential Fourier Series over an interval $0 < t < T$ is given by $x(t) = \sum (4/(1+jn))exp(j(10\pi nt - n\pi/9))$ for all n. Find:
   a. The fundamental frequency.
   b. The average value.
   c. The amplitude and the phase of the third harmonic.
   d. The average power contained in the first three harmonics.

4. Consider the signal $x(t) = cos(100\pi t)sin(200\pi t) + cos^2(150\pi t)$. Find:
   a. The fundamental frequency.
   b. The coefficients a_n, b_n, and X_n for the signal.
   c. The average power of $x(t)$.

5. Using the integral form, find the Fourier transform of the signal $x(t) = 4.exp(-2.t.sgn(t))$.

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Q1 a. $x(t) = 5e^{-(t+0.1)^2} = 5e^{-\pi\left(\frac{t}{\sqrt{\pi}}+0.1\right)^2}$

$X(f) = \frac{5}{(\sqrt{\pi})^{-1}} e^{j2\pi(0.1)f} e^{-\pi\left(\frac{\pi}{\sqrt{\pi}}f\right)^2}$

$= 5\sqrt{\pi}\, e^{j0.2\pi f} e^{-\pi^2 f^2}$

b. $y(t) = 4\sin(50\pi t)\cos(500\pi t) - 2\sin(550\pi t) + 2\sin(450\pi t)$

$Y(f) = j[\delta(f+275) - \delta(f-275)] - j[\delta(f+225) - \delta(f-225)]$

c. $z(t) = \frac{4}{(1-j\pi t)^2} = \frac{4\times 4}{(2+j2\pi(-t))^2}$

$Z(f) = 16 f e^{-2f} u(f)$

Q2 a. $X(f) = 4e^{-4f}u(f) =$

$x(t) = \frac{4}{4 - j2\pi t}$

b. $Y(f) = \left(\frac{-j}{\pi f}\right) e^{j0.2\pi f} = \left(\frac{-j}{\pi f}\right) e^{j2\pi(0.1)f}$

$y(t) = \text{sgn}(t+0.1)$

Q3 $x(t) = \sum \frac{4}{1+jn} e^{-jn\pi/9} e^{j2\pi(5)nt}$

$X_n = \frac{4}{1+jn} e^{-jn\pi/9}$

a. $f_0 = 5$

b. $X_0 = 4$ the average value

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Continue

Q.3 c. $X_3 = \frac{4}{1+j3} e^{-j\frac{\pi}{3}} = \frac{4}{\sqrt{10}} e^{-j\left(\frac{\pi}{3} + \tan^{-1}(3)\right)}$

d.

$|X_1| = \frac{4}{\sqrt{2}}$, $|X_2| = \frac{4}{\sqrt{5}}$, $|X_3| = \frac{4}{\sqrt{10}}$.

$P_{avg} = |X_0|^2 + 2\left[|X_1|^2 + |X_2|^2 + |X_3|^2\right]$.

$= 16 + 2\left[\frac{16}{2} + \frac{16}{5} + \frac{16}{10}\right] = 16 + 32\left[5 + 2 + 1\right]\frac{1}{10}$

$= 16 + 32 \times \frac{8}{10}$

Q.4 $x(t) = \cos(100\pi t)\sin(200\pi t) + \cos^2(150\pi t)$

$= \frac{1}{2} + \frac{1}{2}\sin(100\pi t) + \frac{1}{2}\cos(300\pi t) + \frac{1}{2}\sin(300\pi t)$

a. $f = 50$ Hz

b. $a_0 = \frac{1}{2}$, $b_1 = \frac{1}{2}$, $a_3 = \frac{1}{2}$, $b_3 = \frac{1}{2}$

$X_0 = \frac{1}{2}$, $X_1 = -j\frac{1}{4}$, $X_3 = \frac{1}{4} - j\frac{1}{4}$.

c. $P_{avg} = |X_0|^2 + 2\left[|X|^2 + |X_3|^2\right] = \frac{1}{4} + 2\left[\frac{1}{16} + \frac{1}{16} + \frac{1}{16}\right]$

$= \frac{10}{16} = 1.75$ W

Q.5 $x(t) = 4e^{-2t\,\mathrm{sgn}(t)} = \begin{cases} 4e^{2t} & t < 0 \\ 4e^{-2t} & t > 0 \end{cases}$

$X(f) = \int_{-\infty}^{0} 4e^{(2-j2\pi f)t}\,dt + \int_{0}^{\infty} 4e^{-(2+j2\pi f)t}\,dt$

$= \frac{+4}{(2-j2\pi f)} + \frac{4}{(2+j2\pi f)} = 4 \cdot \frac{4}{4 + (2\pi f)^2}$

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Al Isra Private University
Faculty of Engineering
Electrical Engineering Department

Student Name:_____
Student Number:_____

First Semester: 2005/2006
Second Exam

Course Name:	Systems & Signals Processing	Date:	Dec. 21, 2005
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $y(t) = 4.\cos(200t).\mathrm{sinc}(4t - 8)$.
   b. $z(t) = 8/|4 + j\pi t|^2$.

2. Using Table 4-1 and Table 4-2, find the Inverse Fourier Transform of:
   a. $X(f) = 8.\exp(j5\pi f)/|4 + j\pi f|^2$.
   b. $W(f) = 8/(4 + j200\pi f)^2$.

3. Use the differentiation theorem to find the Fourier transform of the signal $x(t)$ defined by $x(t) = t$, for $|t| \leq 2$, and $x(t) = 2.\mathrm{sgn}(t)$, for $|t| > 2$.

4. Consider a periodic signal $x(t)$ where $x(t)$ in the fundamental period is given by $x(t) = 2t$, for $0 < t < 2$, $x(t) = 2(2 - t)$, and the period $T = 4$. Using Tables 3-1 and 3-2, find the Fourier series complex coefficients X_n then find the power contained in the first three harmonics.

5. Consider a fixed linear system with amplitude response $3.\Pi[w/600\pi] + 3.\Pi[w/400\pi]$ and phase response $-j0.002\,w$ for $|w| \leq 250\pi$, 0.5π for $w < -250\pi$, and -0.5π for $w > 250\pi$. Find the output and determine the type of distortion for the following inputs:
   a. $w(t) = \cos(250\pi t) + 4\cos(275\pi t)$.
   b. $y(t) = \cos(150\pi t) + \cos(250\pi t)$.

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1.1 a $y(t) = 4\cos(200t)\,\mathrm{sinc}(4t-8) = 4\cos\left(2\tfrac{100}{\pi}\pi t\right)\mathrm{sinc}(4(t-2))$

$$Y(f) = \frac{1}{2}\left[e^{-j4\pi\left(f-\frac{100}{\pi}\right)}\,\pi\left(\frac{f-\frac{100}{\pi}}{4}\right) + e^{-j4\pi\left(f+\frac{100}{\pi}\right)}\,\pi\left(\frac{f+\frac{100}{\pi}}{4}\right)\right]$$

b. $$z(t) = \frac{8}{|4+j\pi t|^2} = \frac{8}{16+(\pi t)^2} = \frac{4\times 8}{4\times 16+(2\pi t)^2}$$

$$= 2\,\frac{2\times 8}{8^2+(2\pi t)^2}$$

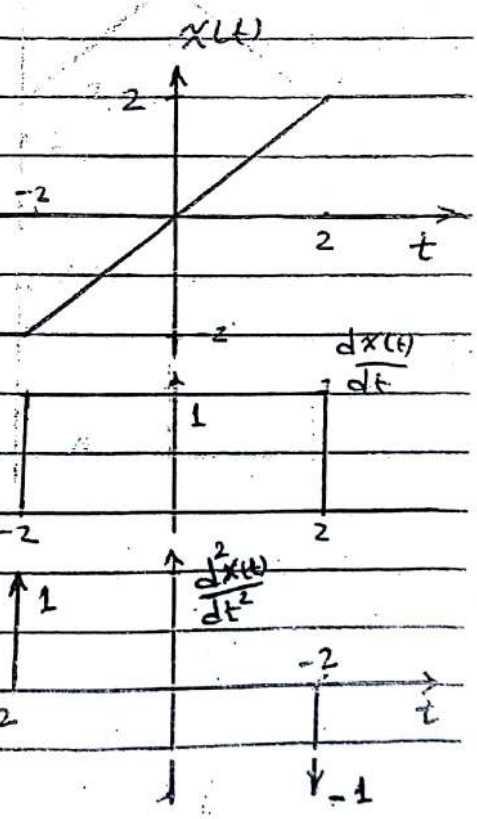
$$Z(f) = 2\,e^{-8|f|}$$

2 a $$X(f) = \frac{8.e^{j5\pi f}}{16+(\pi f)^2} = 2.\frac{2\times 8}{8^2+(2\pi f)^2}\,e^{j2\pi(2.5)f}$$

$$x(t) = 2\,e^{-8|t+2.5|}$$

b. $$W(f) = \frac{8}{(4+j200\pi f)^2} = \frac{8.}{(100)^2[0.04+j2\pi f]^2}$$

$$w(t) = 8\times 10^{-4}\,t\,e^{-0.04t}\,u(t)$$



3 $$\frac{d^2x}{dt^2} = -[\delta(t-2)+\delta(t+2)] = y(t)$$

$$Y(f) = -j2\sin(4\pi f)$$

$$Y(f) = (j2\pi f)^2 X(f)$$

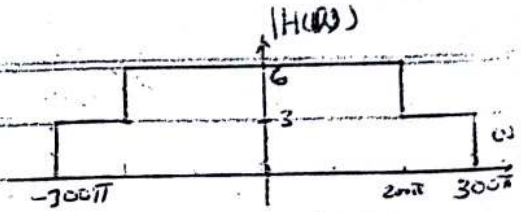
$$X(f) = j\frac{0.5}{(\pi f)^2}\sin(4\pi f)$$

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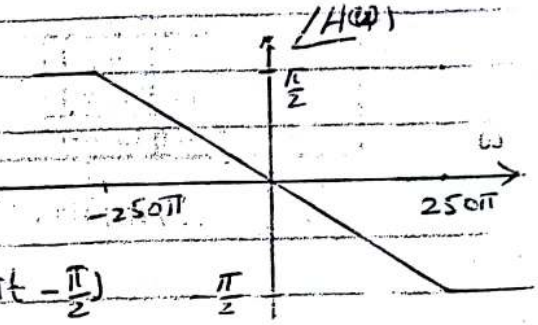
5 a. $u(t) = \cos(250\pi t + 4\cos(2750\pi t)$

$y(t) = 3\cos(250\pi t - \frac{\pi}{2}) + 12\cos($

$275\pi t - \frac{\pi}{2})$.



b There is no distortion



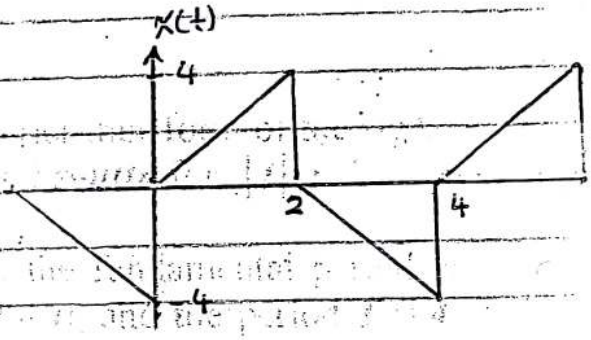
b. $y(t) = \cos(150\pi t) + \cos(250\pi t)$

$y_o(t) = 6\cos(150\pi t - 0.3\pi) + 3\cos(250\pi t - \frac{\pi}{2})$

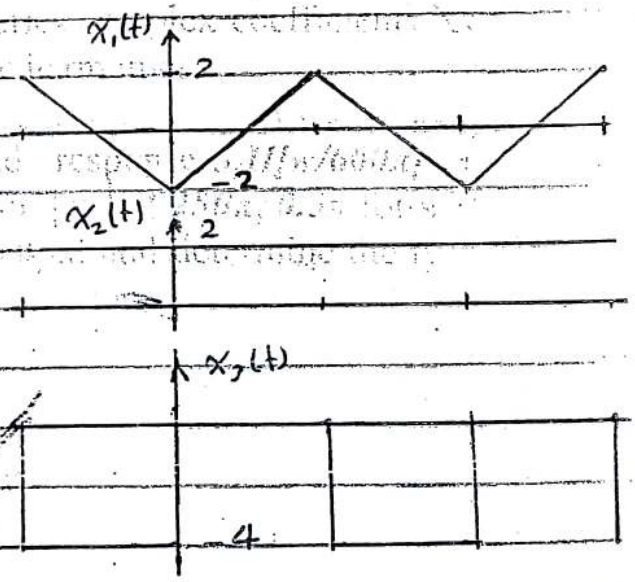
Amplitude distortion

4

$$x(t) = \begin{cases} 2t & 0 < t < 2 \\ 2(2-t) & 2 < t < 4 \end{cases}$$



$x(t) = x_1(t) + x_2(t) + x_3(t)$.



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10

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Faculty of Engineering
Electrical Engineering Department

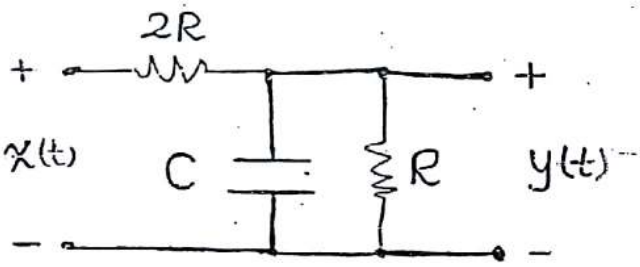
Student Name:_____
Student Number:_____

Second Semester: 2004/2005
Second Exam

Course Name:	Systems & Signals Processing	Date:	May 16, 2005
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

Note: Answer only 4 problems of the 5 below.

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $x(t) = 5.exp(-4t + 1).u(t)$.
   b. $y(t) = 4.sin(50\pi t).cos(500\pi t)$.
   c. $z(t) = 4/|2 + j\pi t|^2$.
2. The complex exponential Fourier Series over an interval $0 < t < T$ is given by $x(t) = \sum (4/(1 + jn))exp(j(5\pi nt - n\pi/8))$ for all n. Find:
   a. The average value. b. The fundamental frequency.
   c. The amplitude and the phase of the third harmonic.
   d. The average power contained in the first three harmonics.
3. Consider a fixed linear system with amplitude response $2.\Pi[w/600] + 2.\Pi[w/300]$ and phase response $-0.0025\pi\, w$ for $|w| < 200$, 0.5π for $w < -200$, and -0.5π for $w > 200$. Find the output and determine the type of distortion for the following inputs:
   a. $x_1(t) = cos(100t) + 2\, cos(200t)$. b. $x_2(t) = cos(180t) + cos(250t)$.
4. Using the integral form, find the inverse Fourier transform of the signal $X(f) = 4.exp(-2.f.sgn(f))$.
5. Using the Fourier transform theory, find the impulse response $h(t)$ of the circuit shown in Fig. 1.



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Q1 a. $x(t) = 5e^{-4t+1}u(t) = 5e^{-4(t+0.25)}u(t)$

(5) $X(f) = 5\dfrac{e^{-j2\pi f(0.25)}}{4+j2\pi f}$

b. $x(t) = 4\sin(50\pi t)\cos(500\pi t)$

$= -j[\delta(f-25) - \delta(f+25)] * [\delta(f-250) + \delta(f+250)]$

$= -j[\delta(f-275) - \delta(f+275) - \delta(f-225) + \delta(f+22$

c. $Z(t) = \dfrac{4}{|2+j\pi t|^2} = \dfrac{4}{4+(\pi t)^2} = 2\cdot\dfrac{2(4)}{4^2+(2\pi t)^2}$

$Z(f) = 2e^{-4|-f|} = 2e^{-4|f|}$

Q2 $x(t) = \sum \dfrac{4}{1+jn} e^{-j\frac{n\pi}{8}} e^{j5\pi nt}$

(6) $X_n = \dfrac{4}{1+jn} e^{-j\frac{n\pi}{8}}$

1a. The average value $= X_0 = 4$,

2b. $X_3 = \dfrac{4}{1+j3} e^{-j\frac{3\pi}{8}}$ $\quad |X_3| = \dfrac{4}{\sqrt{1+9}} = \dfrac{4}{\sqrt{10}}$

$\theta_3 = -\dfrac{3\pi}{8} - \tan^{-1}(3) =$

2d. $|X_1| = \dfrac{4}{\sqrt{1+1}} = \dfrac{4}{\sqrt{2}}$, $X_2 = \dfrac{4}{\sqrt{5}}$

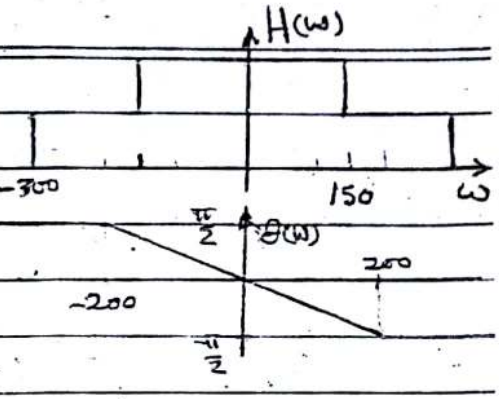
$P_{av} = |X_0|^2 + 2[|X_1|^2 + |X_2|^2 + |X_3|^2 = 16 + 2\left[\dfrac{16}{2} + \dfrac{16}{5} + \dfrac{16}{10}\right.$

$= 16 + 2\times 16\times\dfrac{8}{10} =$

1b. $\dfrac{D}{h} = \dfrac{5\pi}{2\pi} = 2.5$ rad.

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13 $|H(f)| = 2\Pi\left(\frac{\omega}{600}\right) + 2\Pi\left(\frac{\omega}{300}\right)$



5) $\theta(f) = \begin{cases} -0.0025\pi w & |w| \leq 200 \\ 0.5\pi & w < -200 \\ -0.5\pi & w > 200 \end{cases}$

a $x_1(t) = \cos(100t) + 2\cos(200t)$

$y_1(t) = 4\cos(100t - 0.25\pi) + 4\cos(200t - 0.5\pi)$

amplitude distortion

b. $x_2(t) = \cos(180t) + \cos(250t)$

$y_2(t) = 2\cos(180t - 0.45\pi) + 2\cos(250\pi t - \frac{\pi}{2})$

Phase distortion

24 $X(f) = 4e^{-2f\,sgn(f)}$

6

$$x(t) = \int_{-\infty}^{0} 4e^{2f} e^{j2\pi ft}\,df + \int_{0}^{\infty} 4e^{-2f} e^{j2\pi ft}\,df$$

$$= \int_{-\infty}^{0} 4e^{(2+j2\pi t)f}\,df + \int_{0}^{\infty} 4e^{-(2-j2\pi t)f}\,df$$

$$= \frac{4}{(2+j2\pi t)} + \frac{4}{(2-j2\pi t)} = \frac{16}{4+(2\pi t)^2}$$

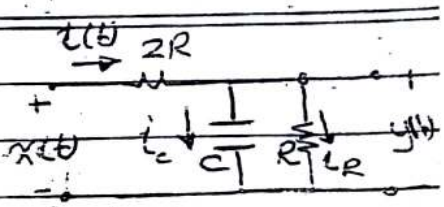
$$= 4\,\frac{2(2)}{(2)^2 + (2\pi t)^2}$$

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15

$i_R(t) = y(t)/R$

$i_c(t) = C\frac{dy}{dt}$



$i(t) = i_R(t) + i_c(t)$

$-x(t) + 2R\,i(t) + y(t) = 0$

$2RC\frac{dy}{dt} + 3y(t) = x(t)$

$Y(f)[3 + 2RC\,j2\pi f] = X(f)$

$H(f) = \frac{Y(f)}{X(f)} = \frac{1}{[3 + 2RC\,j2\pi f]}$

$= \frac{1/2RC}{\frac{3}{2RC} + j2\pi f}$

$h(t) = \frac{1}{2RC} \cdot e^{-\frac{3t}{2RC}} u(t)$.

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Al Isra Private University
Faculty of Engineering
Electrical Engineering Department

Student Name:_____ 14
Student Number:_____

First Semester: 2004/2005
Second Exam

Course Name:	Systems & Signals Processing	Date:	Jan. 3, 2005
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $x(t) = 4.exp(-4t).sin(400\pi t)u(t)$.
   b. $y(t) = (1/t).cos(200\pi t)$.
   c. $z(t) = 8/(4 + j200\pi t)^2$.

2. Using Table 4-1 and Table 4-2, find the Fourier Transform Inverse of:
   a. $X(f) = 3.sinc(4f - 8)$.
   b. $Y(f) = 5.exp(j20\pi f).\Lambda(0.2f)$.
   c. $Z(f) = 8/|4 + j8\pi f|^2$.

3. Use the differentiation theorem to find the Fourier transform of the the signal $x(t) = r(t + 2) - 2.r(t + 1) + 2.r(t - 1) - r(t - 2)$.

4. Consider a fixed linear system with amplitude response $2.\Pi[w/600\pi] + 4.\Pi[w/400\pi]$ and phase response $-j0.002\ w$ for $|w| < 250\pi$, 0.5π for $w < -250\pi$, and -0.5π for $w > 250\pi$. Find the output and determine the type of distortion for the following inputs:
   a. $w(t) = cos(250\pi t) + 4\ cos(275\pi t)$.
   b. $x(t) = cos(50\pi t) + 2\ cos(150\pi t)$.
   c. $y(t) = cos(150\pi t) + cos(250\pi t)$.

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Q1 a. $x(t) = 4e^{-4t} \sin(400\pi t)\, u(t)$ [6]

$$X(f) = \frac{4}{4 + j2\pi f} * \frac{1}{2j}(\delta(f-200) - \delta(f+200))$$

$$= \frac{+j2}{4 + j2\pi(f+200)} - \frac{j2}{4 + j2\pi(f-200)}$$

b. $y(t) = \frac{1}{t} \cdot \cos(200\pi t)$

$$Y(f) = -j\pi \operatorname{sgn}(f) * \frac{1}{2}[\delta(f-100) + \delta(f+100)]$$

$$= -j\frac{\pi}{2}[\operatorname{sgn}(f-100) + \operatorname{sgn}(f+100)]$$

c. $z(t) = \frac{8}{(4 + j200\pi t)^2} = \frac{8}{(100)^2[0.04 + j2\pi t]^2}$

$$Z(f) = \frac{8}{(100)^2} \cdot (-f)\, e^{0.04f}\, u(-f).$$

Q2 a. $X(f) = 3\,\text{sinc}(4(f-2)) = \frac{3}{4}(4\,\text{sinc}(4(f-2)))$ [6]

$$x(t) = \frac{3}{4} e^{j4\pi t}\, \Pi\left(\frac{t}{4}\right).$$

b. $Y(f) = 5.e^{j2\pi(10)f}\, \Lambda(0.2f) = 5\, e^{j2\pi(10)f}\, \Lambda\left(\frac{f}{5}\right)$

$$y(t) = 5.5\,\text{sinc}^2(5(t+10)) = 25\,\text{sinc}^2(5(t+10)).$$

c. $Z(f) = \frac{8}{|4 + j8\pi f|^2} = \frac{8}{(4)^2 + (4)^2(2\pi f)^2}$

$$= \frac{8}{16} \frac{1}{(1 + (2\pi f)^2)} = \frac{1}{4} \frac{2}{(1 + (2\pi f)^2)}$$

$$z(t) = 0.25\, e^{-|t|}$$

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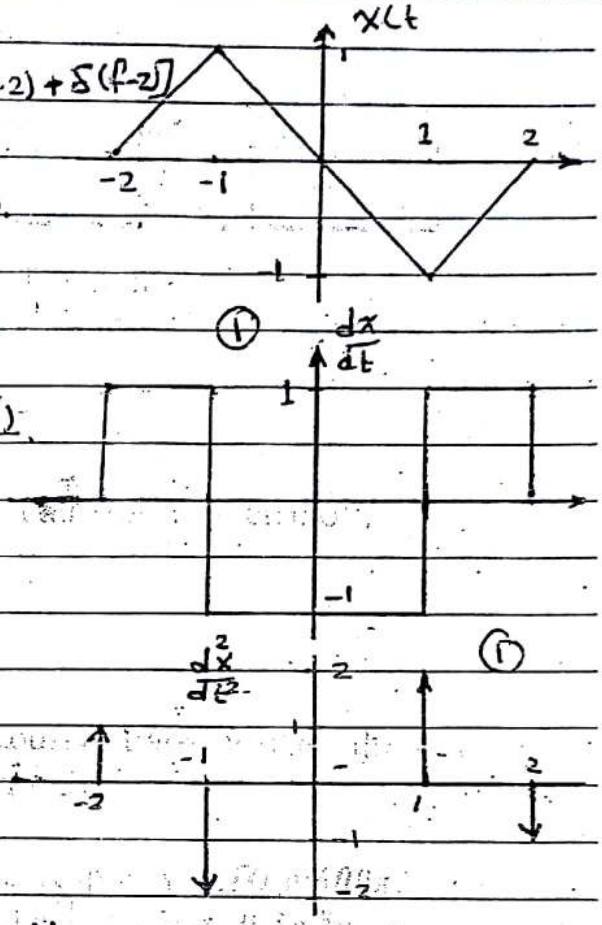
(5) $\frac{d^2x}{dt^2} = 2[\delta(t-1) - \delta(t+1)] + 1[\delta(f+2) + \delta(f-2)]$

$F\{\frac{d^2x}{dt^2}\} = j4\sin(2\pi f) + j2\sin(4\pi f)$.

$= (j2\pi f)^2 \; X(f)$.

$X(f) = -j\frac{\sin(2\pi f)}{(\pi f)^2} + j\frac{1}{2}\cdot\frac{\sin(4\pi f)}{(\pi f)^2}$

$X(f) = X(-f)$



Q.4 a. $x(t) = \cos(250\pi t) + 4\cos(275\pi t)$

10 $w_o(t) = 2\cos(250\pi t - \frac{\pi}{2}) + 8\cos(275\pi t - \pi/2)$.

No distortion.

b. $x(t) = \cos(50\pi t) + 2\cos(150\pi t)$

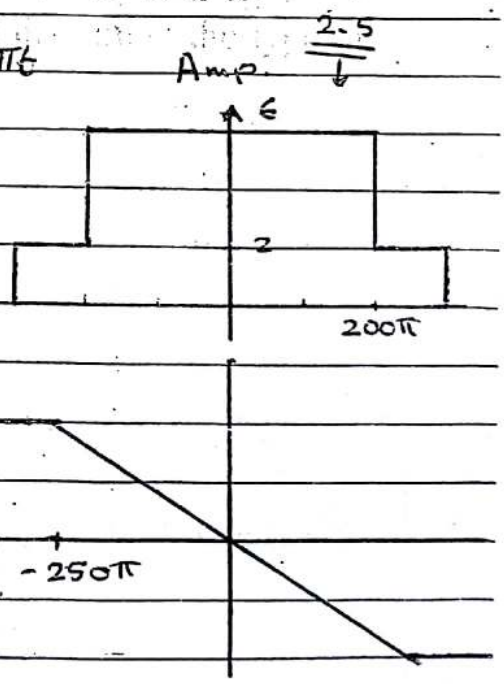
$x_o(t) = 6\cos(50\pi t - 0.1\pi) + 12\cos(150\pi t - 0.3\pi)$.

No distortion.

c. $y(t) = \cos(150\pi t) + \cos(250\pi t)$

$y_o(t) = 6\cos(150\pi t - 0.3\pi) + 2\cos(250\pi t - \frac{\pi}{2})$

Amplitude distortion.



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Al-Isra university
Faculty of Engineering
Electrical Engineering Department
Eng. Juma Abu-Ghalune
August 19, 2004
Second Exam
Signal Processing and Systems

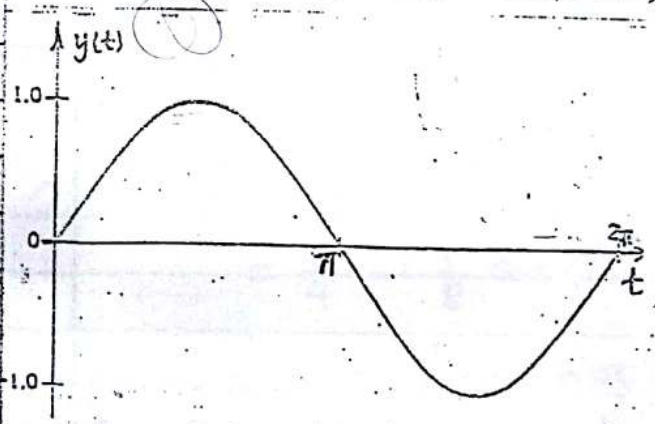
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1. **[7 marks]** The complex exponential Fourier Series of a periodic signal with period T is given by $x(t) = \sum (4\text{sinc}(n/4))\exp(-j20\,\pi nt)$ for all n. Find:
   a. The average value. b. The fundamental frequency.
   c. The amplitude and the phase of the third harmonic.
   d. X_n, a_n, and b_n for $n < 4$.
   e. The average power contained in the first three harmonics.

2. **[8 marks]** Consider the signal signal $x(t) = \cos^2(100\pi t)\sin^2(150\pi t)$. Find:
   a. The complex coefficients X_n.
   b. The fundamental frequency.
   c. The average power of the signal.

3. **[10 marks]** Using Table 4-1 and Table 4-2, find:
   A. The Inverse Fourier Transform of:
      a. $X(f) = \exp(j4f)\Lambda(0.5f)$.
      b. $Y(f) = 4\cos(10\,\pi f) / |4 + j2\pi f|^2$.
   B. The Fourier Transform of the signals shown in Fig. 1.

4. **[Bonus]** Consider a fixed linear system with amplitude response $A + A.\Pi[f/200]$ and phase response $-0.01\pi f$ for $|f| < 50$, 0.5π for $f < -50$, and -0.5π for $f > 50$. Find the output and determine the type of distortion for the following inputs:
   a. $y(t) = \cos(120\pi t) + \cos(220\pi t)$. b. $w(t) = \cos(80\pi t) + 4\cos(160\pi t)$.



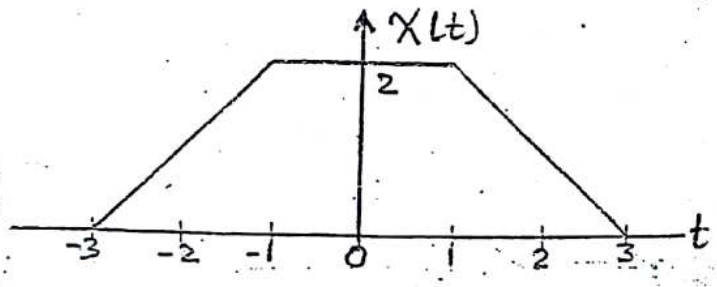


Fig. 1

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1 $x(t) = \sum_{n=-\infty}^{\infty} 4\,\text{sinc}\left(\frac{n}{4}\right) e^{-j20\pi nt}$ $\quad 0 \le t \le T.$

$X_n = 4\,\text{sinc}\left(\frac{n}{4}\right)$

1 a. The average value $= X_0 = 4$.

1 b. $20\pi = 2\pi f_0 \rightarrow f_0 = 10$ Hz.

1.5 c. $X_3 = 4\,\text{sinc}\left(\frac{3}{4}\right) = 4\,\frac{\sin(3\pi/4)}{3\pi/4} = 1.2$

$|X_3| = 1.2, \quad \theta_3 = 0.$

2.5 d. $X_n = \frac{1}{2} a_n$ for all n since X_n is real.

$X_0 = 4, \quad X_1 = 4\,\text{sinc}\left(\frac{1}{4}\right) = 4\,\frac{\sin(\frac{1}{4}\pi)}{(1/4)\pi} = 3.6.$

$X_2 = 4\,\text{sinc}\left(\frac{1}{2}\right) = 2.55, \quad X_3 = 4\,\text{sinc}\left(\frac{3}{4}\right) = 1.2$

$X_{-n} = X_n, \quad a_0 = 4, \; a_1 = 7.2, \; a_2 = 5.1, \; a_3 = 2.4.$

1 e. $P = X_0^2 + 2\sum_{n=1}^{3} |X_n|^2 = 57.8$

2 $x(t) = \cos^2(100\pi t)\sin^2(150\pi t) = \frac{1}{4}[1+\cos(200\pi t)][1-\cos(300\pi t)]$

2 $= \frac{1}{4} - \frac{1}{8}\cos(100\pi t) + \frac{1}{4}\cos(200\pi t) - \frac{1}{4}\cos(300\pi t)$

$- \frac{1}{8}\cos(500\pi t)$

3 a. $a_0 = \frac{1}{4}, \; a_1 = -\frac{1}{8}, \; a_2 = \frac{1}{4}, \; a_3 = -\frac{1}{4}, \; a_5 = -\frac{1}{8}.$

$X_0 = \frac{1}{4}, \quad X_1 = -\frac{1}{16}, \quad X_2 = \frac{1}{8}, \quad X_3 = -\frac{1}{8}$

$X_5 = -\frac{1}{16}.$

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2 b. The fundamental frequency = 50.

1

2 c. $P = X_0^2 + 2\sum_{n=1}^{\infty} |X_n|^2 = (\frac{1}{4})^2 + 2[(\frac{1}{16})^2 + (\frac{1}{8})^2 + (\frac{1}{8})^2 + (\frac{1}{16})^2]$

$= \frac{9}{64}$.

3 A. a. $X(f) = e^{j4f}\,\Lambda(0.5f) = \frac{1}{1.2} e^{-j2\pi f(\frac{2}{\pi})} \cdot 2\Lambda(\frac{f}{2})$.

$x(t) = 2\,\text{sinc}^2(2(t+\frac{2}{\pi}))$

b. $Y(f) = \frac{4\cos(2\pi f(5))}{16 + (2\pi f)^2} = \frac{1}{2}\cos(2\pi(5)f)\,\frac{2(4)}{(4)^2 + (2\pi f)^2}$.

$y(t) = \frac{1}{4}[\delta(f+5) + \delta(f-5)] * e^{-4|t|}$

$= \frac{1}{4}[e^{-4|t+5|} + e^{-4|t-5|}]$.

B. $y(t) = \Pi(\frac{t-\pi}{2\pi})\sin(t)$

$Y(f) = 2\pi e^{-j2\pi^2 f}\,\text{sinc}(2\pi f) * \frac{1}{2j}[\delta(f - \frac{1}{2\pi}) + \delta(f + \frac{1}{2\pi})]$

$= -j\pi\, e^{-j2\pi^2(f-\frac{1}{2\pi})}\,\text{sinc}(2\pi(f-\frac{1}{2\pi})) +$

$j\pi\, e^{-j2\pi^2(f+\frac{1}{2\pi})}\,\text{sinc}(2\pi(f+\frac{1}{2\pi}))$.

$x(t) = 3\Lambda(\frac{t}{3}) - \Lambda(t)$.

$X(f) = 9\,\text{sinc}^2(3f) - \text{sinc}^2(f)$.

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Al-Isra university
Faculty of Engineering
Electrical Engineering Department
Eng. Juma Abu-Ghalune
Jan. 5, 2004
Second Exam
Signals and Systems Analysis

1. Find the Fourier Transform of $x(t) = exp(-4t).u(t - 2)$ using the integral form.

2. The complex exponential Fourier Series over an interval $0 < t < T$ is $x(t) = \sum (4/(1 + j\pi n))exp(-j20nt/4)$ for all n. Find:
   a. The average value. b. The fundamental frequency.
   c. The amplitude and the phase of the third harmonic.

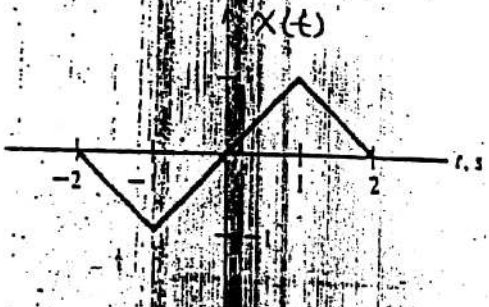
3. Using Table 4-1 and Table 4-2, find the Fourier Transform inverse of:
   a. $X(f) = (1/f)exp(j20\pi f)$.. b. $Y(f) = 4/|2 + j\pi f|^2$.
   c. $Z(f) = 4\pi f.sinc^2(2f)$.

4. Use Parseval's theorem to find the average power for the signal $x(t) = 4.sin^2(150\pi t).cos(200\pi t)$.

5. Consider a fixed linear system with amplitude response $2A - A.\Pi[f/200]$ and phase response $-0.01\pi f$ for $|f| < 50$, 0.5π for $f < -50$, and -0.5π for $f > 50$. Find the output and determine the type of distortion for the following inputs:
   a. $y(t) = cos(120\pi t) + cos(220\pi t)$. b. $w(t) = cos(80\pi t) + 4\,cos(160\pi t)$.

6. Use the differentiation theorem to find the Fourier Transform of the signal shown in Fig. 1.

Fig. 1



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Q.1 $x(t) = e^{-4t} u(t-2)$

$$X(f) = \int_2^{\infty} e^{-(j2\pi f + 4)t}\, dt = \int_{+2}^{\infty} e^{-t(4+j2\pi f)}\, dt$$

$$= \frac{-1}{(4+j2\pi f)} e^{-t(4+j2\pi f)} \Big]_{+2}^{\infty}$$

$$= \frac{1}{(4+j2\pi f)} e^{-2(4+j2\pi f)} = \frac{e^{-8}}{(4+j2\pi f)} e^{-j4\pi f}$$

Q.2 $x(t) = \sum_{n=-\infty}^{\infty} \frac{4}{(1+j\pi n)} \cdot e^{j5nt} \qquad 0 < t < T$

a. $n = 0 \rightarrow X_0 = 4$: average value

b. $n = 1 \rightarrow 2\pi f_0 = 5 \rightarrow f_0 = \frac{5}{2\pi}$

c. $X_3 = \frac{4}{(1+j3\pi)} = \frac{4}{\sqrt{1+9\pi^2}} \angle -\tan^{-1}(3\pi)$

$$= \frac{4}{\sqrt{1+9\pi^2}} \angle -83.9$$

Q3 a. $X(f) = \frac{1}{f} e^{j20\pi f} \longleftrightarrow j\pi \operatorname{sgn}(t+10)$

Time delay + 14.

b. $Y(f) = \frac{4}{|2+j\pi f|^2} = \frac{4}{4+(\pi f)^2} = \frac{16}{16+(2\pi f)^2} = 2\frac{2(4)}{4^2+(2\pi f)^2}$

$y(t) = 2e^{-4|t|}$

c. $Z(f) = 4\pi f \operatorname{sinc}^2(2f) = -j(j2\pi f)\, 2\operatorname{sinc}^2(2f)$.

$$\frac{d\Lambda\left(\frac{t}{2}\right)}{dt} \quad -j\pi\left(-\frac{1}{2}\right) \quad +j\pi\frac{1}{2}$$

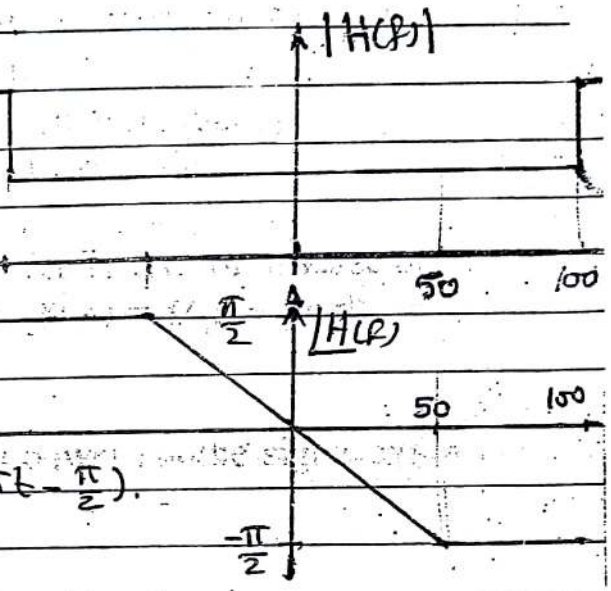
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24 $x(t) = 4\sin^2(150\pi t)\cos(200\pi t) = 2[1-\cos(300\pi t)]\cos(200\pi t)$

$= -\cos(100\pi t) + 2\cos(200\pi t) + \cos(500\pi t)$

$X_1 = \frac{-1}{2}, \quad X_2 = 1, \quad X_5 = \frac{-1}{2}$

$P_{avg} = 2\left[\frac{1}{4} + 1 + \frac{1}{4}\right] = 3$



25 a. $y(t) = \cos(120\pi t) + \cos(220\pi t)$

$y_o(t) = A\cos(120\pi t - \frac{\pi}{2}) + 2A\cos(220\pi t - \frac{\pi}{2})$

Amplitude distortion.

b. $w(t) = \cos(80\pi t) + 4\cos(160\pi t)$

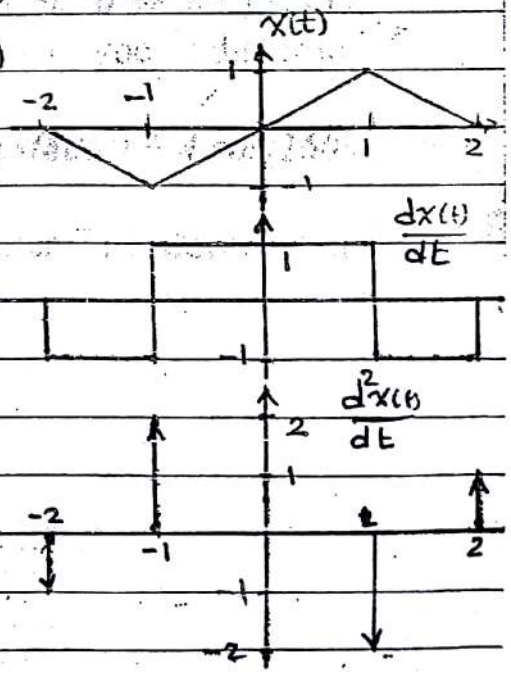
$w_o(t) = A\cos(8\pi t - 0.4\pi) + 4A\cos(160\pi t - \frac{\pi}{2})$.

Phase distortion.

26 $\frac{d^2x(t)}{dt^2} = +\delta(t-2) + \delta(t+2) - 2\delta(t-1) + 2\delta(t+1)$

$F\{\frac{d^2x(t)}{dt^2}\} = e^{-j2\pi f2} + e^{j2\pi f2} - 2e^{-j2\pi f} + 2e^{+j2\pi f}$

$= -j2\sin(4\pi f) - j4\sin(2\pi f)$.



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